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Transcutaneous electrical acupoint stimulation for postoperative nausea and vomiting in patients undergoing craniotomy: A randomized controlled trial

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ABSTRACT

Background: The incidence of nausea and vomiting following craniotomy is high, and pericardium 6 (P6; Neiguan) acupoint stimulation is an important strategy for treating postoperative nausea and vomiting (PONV). Here, we aimed to evaluate the efficacy of transcutaneous electrical acupoint stimulation (TEAS) at P6 as an adjunct to antiemetic drugs to prevent PONV after craniotomy.
Materials and methods: This randomized placebo-controlled trial enrolled 120 patients scheduled for craniotomy. The enrolled patients were randomly assigned to a TEAS or sham TEAS group. The incidence of PONV, pain score, and postoperative remedial treatment with antiemetics and analgesics at 0–2, 2–6, and 6–24 h after craniotomy were assessed.
Results: The patient characteristics did not significantly differ between the two groups ($P > 0.05$). During 0–2 and 6–24 h after craniotomy, the incidence of vomiting was not significantly different between the two groups ($P > 0.05$). During 2–6 h, the incidence of vomiting was higher in the sham TEAS group than in the TEAS group (29.3 % vs. 14.0 %, $P = 0.047$). During 0–2 and 2–6 h, the pain scores did not differ significantly between the two groups ($P > 0.05$). During 6–24 h after craniotomy, the pain score was significantly higher in the sham TEAS group than in the TEAS group ($P = 0.001$). The degree of nausea and proportion of patients requiring antiemetic drugs were not significantly different between the two groups in each period ($P > 0.05$).
Conclusion: TEAS at P6 may reduce vomiting incidence and pain scores following craniotomy.

1. Introduction

The incidence of postoperative nausea and vomiting (PONV) following craniotomy can reach as high as 55%–70 % [1]. Severe vomiting after craniotomy may result in intracranial hematoma and hemorrhage, consequently prolonging the hospital stay of patients and reducing their satisfaction with the surgical procedure [2]. The pathophysiology of PONV is complex and largely unknown. However, a recent study has suggested that PONV occurs via multiple peripheral and central mechanisms mediated by various neurotransmitters and receptors [3]. Different antiemetics, including serotonin receptor antagonists, glucocorticoids, anticholinergics, butyrophenones, antihistamines, phenothiazines, and neurokinin 1 receptor antagonists, are widely used in clinical practice for the pharmacological prophylaxis and treatment of PONV [4]. However, some drugs have been reported to cause adverse events (AEs), including excessive sedation, extrapyramidal signs, hallucinations, dysphoria, and hypotension [5,6], which are undesirable in postoperative neurological assessment following craniotomy.

In addition to drug therapy, nonpharmacological approaches can be employed for PONV prophylaxis. A commonly cited example is pericardium 6 (P6; Neiguan) acupoint stimulation. The fourth Consensus Guidelines for the Management of PONV states that P6 stimulation considerably reduces the risk of nausea and vomiting as well as the need for rescue antiemetics compared with sham treatment (evidence A1). According to these guidelines, P6 stimulation is an essential multimodal preventive treatment method for PONV [4,7]. P6 stimulation has been reported to exert antiemetic effects by regulating the excitability of the

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autonomic nervous system (sympathetic and parasympathetic) [8] and the release of substance P and motilin [9].

With the development of enhanced recovery after surgery, emphasis on refining the management of PONV according to the surgical site has increased. However, the effectiveness and safety of P6 stimulation in managing postcraniotomy PONV remain unknown [10]. Furthermore, we previously reported that acupuncture at P6 can reduce vomiting incidence and pain scores following craniotomy [11]. However, patients have reported discomfort and pain associated with acupuncture needling and expressed a desire for a more comfortable and painless form of P6 stimulation; these needs are fulfilled by TEAS, which is a painless stimulation method. A previous study has confirmed the effectiveness of TEAS in preventing postcraniotomy vomiting [12]; in this study, multiple acupoints were used, including Hegu and Zusanli, in addition to TEAS at P6; however, the selection of numerous acupoints for TEAS led to a complex operational method. With the development of wearable devices, we hope to clarify the effectiveness of single-acupoint stimulation at P6 for treating postcraniotomy vomiting. This will lay the foundation for the future development of wearable stimulation devices targeting the P6 acupoint. Thus, to evaluate the efficacy of TEAS at P6 as an adjunct to antiemetic drugs for preventing PONV after craniotomy, we compared the effects of TEAS at P6 and sham TEAS at P6 through a prospective randomized placebo-controlled trial.

2. Materials and methods

2.1. Study design

This randomized blinded controlled trial was registered in the Chinese Clinical Trial Registry in 2013 (registration number: ChiCTR-TRC-13003026). The study protocols were published in trials [13] and were approved by the Ethics Review Board of West China Hospital, Sichuan University (approval number: 2012249).

2.2. Study population

In this prospective randomized controlled trial, patients were screened according to the inclusion and exclusion criteria. Patients were recruited from the Department of Neurosurgery at the West China Hospital, Sichuan University between October 2014 and September 2018. In total, 120 patients were recruited.

2.2.1. Inclusion criteria

Patients who fulfilled the following criteria were included: scheduled for neurosurgery requiring opening of the cranium and dura, aged between 18 and 70 years, American Society of Anesthesiologists physical status classification of I or II, undergoing general anesthesia, no history of PONV or motion sickness, no use of antiemetics and analgesia 24 h before surgery, willingness to participate, no prior experience with TEAS and signed an informed consent form.

2.2.2. Exclusion criteria

Participants who met any of the following criteria were excluded: vomiting 24 h before surgery; pregnant or lactating women; drug or alcohol abusers; recipients of chemotherapy or radiation therapy during the previous 7 days; having a cardiac pacemaker fitted; in the menstruating phase of the menstrual cycle; refusal to accept acupuncture treatment; mental disorder; history of epilepsy and still on antiepileptic medicine; unconscious before the surgery; cannot communicate normally; undergoing ventricle or brainstem surgery; poorly controlled diabetes mellitus (fasting plasma glucose greater than 12 mmol/L); and serious systemic disease (acquired immune deficiency syndrome or sepsis).

2.2.3. Withdrawal criteria

The withdrawal criteria for the patients were as follows: death; waking more than 2 h after surgery; tracheal intubation; persistent coma; cognitive impairment; vomiting caused by intracranial hypertension; and further surgery or transfer to the intensive care unit due to disease aggravation.

2.3. Anesthesia and postoperative analgesia

All patients were administered general anesthesia with endotracheal intubation. Blood pressure, heart rate, pulse oximetry, and end-tidal CO_2 levels were routinely monitored. Thirty minutes prior to the end of the operation, the patients were treated with prophylactic antiemetic drugs (ondansetron injection 8 mg), according to the doctor's advice. After surgery, the patients were continually monitored in the post-anesthesia care unit with continued ventilator support. The tracheal tube was removed after the patient gained consciousness. The time from anesthesia induction to tube removal was recorded.

After the participants returned to the neurosurgery ward, the doctors assessed their severity of PONV and pain to determine whether antiemetic and analgesic medications were needed. The patients with difficulty enduring nausea and vomiting received an intramuscular injection of 10 mg metoclopramide. Patients with a visual analog scale (VAS) pain score of >6 received an intramuscular injection of 10 mg tramadol. The degree of PONV and postoperative pain was recorded at the time of medication administration.

2.4. Randomization and blinding

Patients were randomized to either the TEAS or sham TEAS group using a computer-generated random number list at a ratio of 1:1. Allocation concealment was achieved by enclosing the assignments in sealed, opaque, sequentially numbered envelopes, which were opened only after confirming eligibility and providing informed consent. Patient allocation was performed by a clinical assistant who was trained with the institutional review board policies. All patients were unaware of the group to which they were assigned. Outcome assessors, data collectors, and statisticians were also blinded to the group allocation.

2.5. Interventions

In the TEAS group, after transferring the patients from the operating room to the anesthesia resuscitation room, the intervention was administered as the patient gained consciousness. The patients' forearms were cleaned with 75 % alcohol swabs, and self-adhesive electrodes were placed bilaterally over the P6 acupoints. The SDZ-V nerve and muscle stimulators (Hwato, Jiangsu, China) were used for TEAS. The stimulator lasted for 30 min with a dilatational wave of 2/100 Hz, and the intensity varied until the patients continuously felt a de qi sensation [14,15]. At the end of the experiment, all self-adhesive electrodes were removed.

In the sham TEAS group, the patients had self-adhesive electrodes placed bilaterally over the P6 acupoints and connected to the SDZ-V nerve and muscle stimulators (Hwato, Jiangsu, China) for 30 min. Unlike the TEAS group, special wires were used in the sham TEAS group; the copper wires within the unique wire were deliberately severed to prevent the generation of electrical stimulation when activating the stimulators. However, the stimulators still exhibited the same operational power indicator and emitted the same sound. To ensure that patients were blinded to the interventions, each patient was told that a special acupoint stimulation, which cannot be felt via human sensory perception, was used for the treatment. After 30 min, all self-adhesive electrodes were removed, similar to that in the TEAS group (Table 1).

2.6. Measures

The entire study period was 24 h after surgery. A separate research nurse who was not involved in patient management recorded the

Table 1
Details of interventions.

Items	TEAS group (n = 57)	Sham TEAS group (n = 58)
Acupoint	pericardium 6	pericardium 6
Stimulation device	SDZ-V nerve and muscle stimulator	SDZ-V nerve and muscle stimulator
Stimulation time	30 min	30 min
Stimulation frequency display	2/100 Hz	2/100 Hz
Countdown alarm sound	yes	yes
Actual current outuput	yes	no
De qi sensation	yes	no

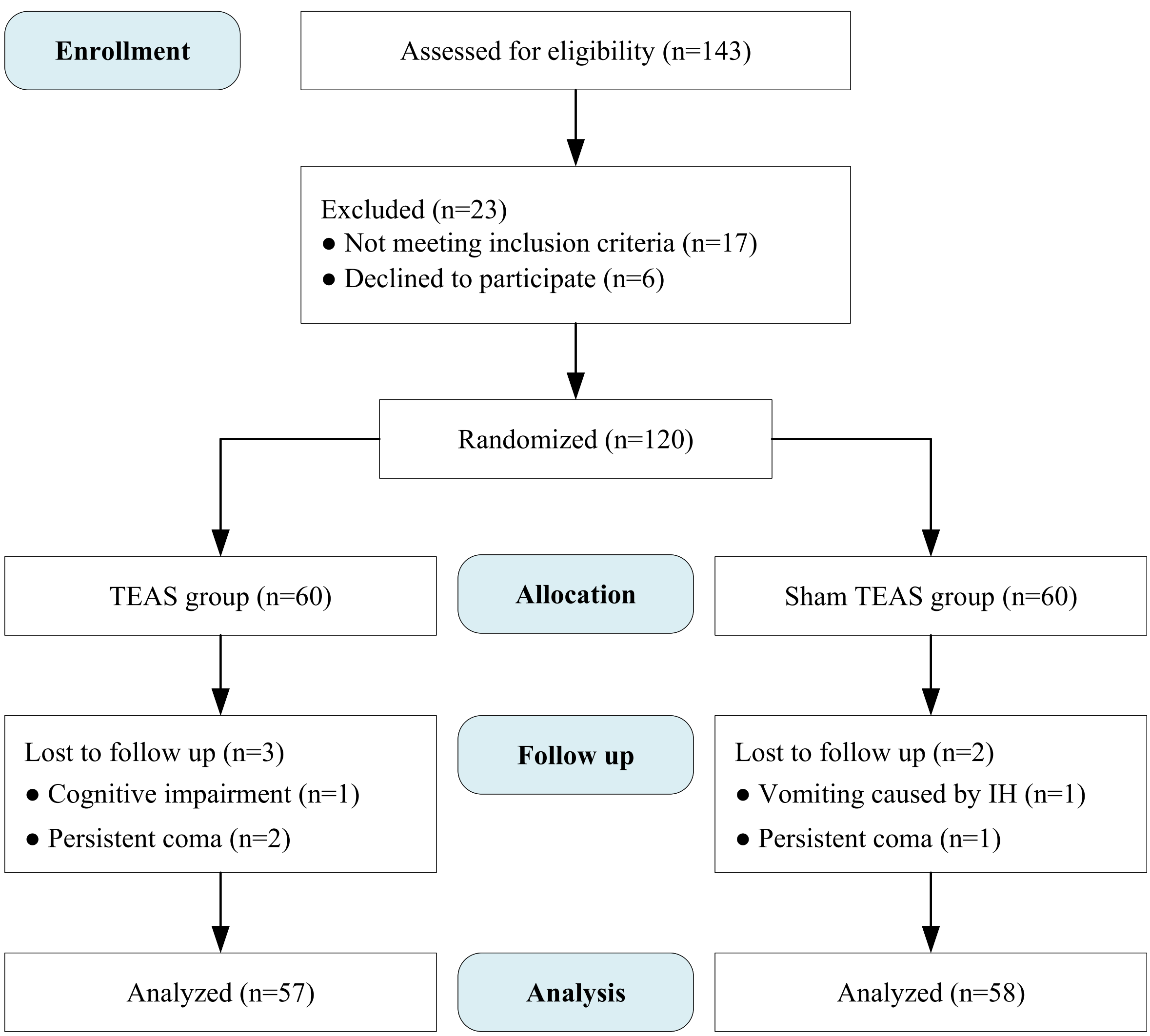
TEAS, transcutaneous electrical acupoint stimulation.

anesthesia time, surgery time, endotracheal intubation time, demographic characteristics, and preoperative data for each patient. The demographic characteristics and preoperative data included age, sex, race, weight, and smoking history. Another blinded observer (nurse) recorded the postoperative data, which included any episodes of vomiting (based on vomiting action or vomiting in the mouth) at 0–2 h, 2–6 h, and 6–24 h postoperatively. The incidence of vomiting within 24 h after craniotomy in the two groups was the main outcome to be measured. The observers evaluated the patients' degree of nausea using the WHO's PONV fourth-class rating scale: 0) no nausea at all; 1) mild nausea or abdominal discomfort without vomiting; 2) evident nausea without vomiting; and 3) extreme nausea and vomiting containing gastric juice, which is uncontrolled by medicine. The pain score was measured using a standard VAS at 0–2, 2–6, and 6–24 h after craniotomy. The total rescue antiemetic and analgesic dosages 0–24 h after craniotomy were recorded. The reasons for withdrawal and TEAS adverse events, including malignant arrhythmia, abnormal blood pressure fluctuations, fainting, serious pain, and local infection, were recorded.

2.7. Sample size calculation and statistical analyses

A previous study reported a PONV incidence of 47 % in patients who had undergone craniotomy with general anesthesia [16]. We hypothesized that TEAS at P6 would reduce the incidence of PONV by 30 % compared with that in the sham TEAS group. To obtain statistical significance and power of 0.05 and 90 %, respectively, 49 participants were required per group (PASS 15.0). With an estimated dropout rate of 20 %, we planned to include 60 participants per group.

All data inputs and statistical analyses were performed at the



TEAS, transcutaneous electrical acupoint stimulation; IH, intracranial hypertension.

Fig. 1. CONSORT flow diagram.

Department of Epidemiology and Hygienic Statistics of the West China Hospital of Sichuan University using SPSS 19.0. Data were checked for normality using Kolmogorov–Smirnov test. Normally distributed variables were presented as mean ± standard deviation and were analyzed using two-sample *t*-test. Non-normally distributed variables were described as median (interquartile range [IQR]) and were compared using Mann–Whitney *U* test. A P-value of <0.05 indicated statistical significance.

3. Results

3.1. Patient characteristics and surgical details

In total, 120 patients were enrolled in the present study, and 5 patients (3 in the TEAS group and 2 in the sham TEAS group) were excluded because they met the withdrawal criteria. Three of these five patients had persistent coma, one exhibited cognitive impairment, and one demonstrated intracranial hypertension-induced vomiting (Fig. 1). Finally, data from 115 patients were analyzed. The patient characteristics did not differ significantly between the TEAS and sham TEAS groups (P > 0.05; Table 2).

Table 2
Demographic data and surgical characteristics.

Characteristic	TEAS group (n = 57)	Sham TEAS group (n = 58)	Statistics	P-value
Age (mean ± SD, years)	44.77 ± 15.74	49.24 ± 14.02	t = −1.609[a]	0.102
Height (mean ± SD, cm)	162.75 ± 7.49	162.64 ± 8.32	t = 0.079[a]	0.917
Weight (mean ± SD, kg)	61.42 ± 10.09	58.19 ± 8.93	t = 1.820[a]	0.069
Anesthesia time (mean ± SD, min)	246.93 ± 95.46	262.10 ± 129.88	t = −0.713[a]	0.958
Operation time (mean ± SD, min)	166.49 ± 88.77	176.64 ± 103.85	t = −0.563[a]	0.801
Dose of sufentanil (mean ± SD, ug)	28.99 ± 10.78	28.16 ± 13.23	t = 0.368[a]	0.350
Infusion amount (mean ± SD, ml)	2348.25 ± 1283.72	2551.72 ± 1227.61	t = −0.869[a]	0.378
Gender			χ2 = 0.010 [b]	0.920
female	31 (54.4)	31 (53.4)		
male	26 (45.6)	27 (46.6)		
History of smoking			χ2 = 0.270 [b]	0.603
no	45 (78.9)	48 (82.8)		
yes	12 (21.1)	10 (17.2)		
History of surgery			χ2 = 2.312 b	0.128
no	42 (73.3)	35 (60.3)		
yes	15 (26.3)	23 (39.7)		
History of migraine			χ2 = 0.708[b]	0.400
no	46 (80.7)	43 (74.1)		
yes	11 (19.3)	15 (25.9)		
History of opioid use			χ2 = 2.582 [b]	0.108
no	45 (78.9)	38 (65.5)		
yes	12 (21.1)	20 (34.5)		
History of motion sickness			χ2 = 0.003 [b]	0.954
no	41 (71.9)	42 (72.4)		
yes	16 (28.1)	16 (27.6)		
Type of surgery			χ2 = 1.769[b]	0.423
supratentorial	35 (61.4)	41 (70.7)		
subtentorial	19 (33.3)	16 (27.6)		
both	3 (5.7)	1 (1.7)		

SD, standard deviation; TEAS, transcutaneous electrical acupoint stimulation.
[a] Two-sample *t*-test.
[b] Chi-squared test; n, number.

3.2. Comparison of PONV between the two groups

Within 0–2 h following craniotomy, 4 patients (7 %) in the TEAS group and 10 patients (17.2 %) in the sham TEAS group experienced vomiting. However, the difference in vomiting incidence between the two groups was not statistically significant (P = 0.095). The vomiting incidence at 2–6 h following craniotomy was significantly higher in the sham TEAS group (17 patients [29.3 %]) than in the TEAS group (8 patients [14.0 %]) (P = 0.47). During 6–24 h following craniotomy, 11 patients (19.3 %) in the TEAS group and 15 patients (25.9 %) in the sham TEAS group experienced vomiting; the incidence of vomiting was not significantly different between the two groups (P = 0.405). During the postoperative periods of 0–2, 2–6, and 6–24 h, both the groups exhibited different degrees of nausea; however, the difference in each period was not statistically significant (P > 0.05; Table 3).

3.3. Comparison of antiemetic and analgesic usage rates and pain scores between the two groups

At 0–6 h following craniotomy, no patient required antiemetic drugs in either group; however, the number of patients who required antiemetic drugs increased between 6 and 24 h following craniotomy. One patient in each group required antiemetic medication. No significant difference was observed in the proportion of patients requiring antiemetic drugs between the two groups (P > 0.05). At 6–24 h following craniotomy, three patients in the TEAS group and six patients in the sham TEAS group received tramadol; there was no significant difference in the tramadol usage rate between the two groups (P = 0.315; Table 4). Moreover, there was no significant difference in the VAS pain scores between the two groups at 0–2 and 2–6 h following craniotomy. Nonetheless, at 6–24 h following craniotomy, the pain score was significantly higher in the sham TEAS group than in the TEAS group (P = 0.001; Table 5). AEs were not observed in any patient during the 0–24-h testing period.

4. Discussion

The main finding of the present study was that TEAS at P6 may prevent vomiting and reduce pain scores but it may not reduce the degree of nausea or the dosage of antiemetic drugs following craniotomy.

We found that TEAS at P6 did not reduce vomiting incidence at 0–2

Table 3
Comparison of PONV between two groups.

Outcome	TEAS group (n = 57)	Sham TEAS group (n = 58)	Statistics	P-value
Vomiting within 0–2 h after operation			χ2 = 2.81[a]	0.095
no	53 (93.0)	48 (82.8)		
yes	4 (7.0)	10 (17.2)		
Vomiting within 2–6 h after operation			χ2 = 3.94[a]	0.047
no	49 (86.0)	41 (70.7)		
yes	8 (14.0)	17 (29.3)		
Vomiting within 6–24 h after operation			χ2 = 0.71[a]	0.405
no	46 (80.7)	43 (74.1)		
yes	11 (19.3)	15 (25.9)		
Score for nausea 0–2 h after operation	0 (0–1)	0 (0–1)	z = 0.671[b]	0.502
Score for nausea 2–6 h after operation	0 (0–1)	0 (0–2)	z = 1.453[b]	0.146
Score for nausea 6–24 h after operation	0 (0–1)	0 (0–1)	z = 0.563[b]	0.574

PONV, postoperative nausea and vomiting; TEAS, transcutaneous electrical acupoint stimulation.
[a] Chi-squared test.
[b] Mann–Whitney *U* test; h, hours; n, number.

Table 4
Comparison of antiemetic and analgetic use between two groups.

Antiemetic and analgetic use in different periods after operation	TEAS group (n = 57)	Sham TEAS group (n = 58)	Statistics	P-value
Use of metoclopramide within 0–2 h			NA	1.000
no	57 (100.0)	58 (100.0)		
yes	0 (0.0)	0 (0.0)		
Use of metoclopramide within 2–6 h			NA	1.000
no	57 (100.0)	58 (100.0)		
yes	0 (0.0)	0 (0.0)		
Use of metoclopramide within 6–24 h			$\chi 2 = 0.001$[a]	0.990
no	56 (98.2)	57 (98.3)		
yes	1 (1.8)	1 (1.7)		
Use of tramadol within 6–24 h			$\chi 2 = 0.45$[a]	0.315
no	54 (94.7)	52 (89.7)		
yes	3 (5.3)	6 (10.3)		

TEAS, transcutaneous electrical acupoint stimulation; NA, not applicable.
[a] Chi-squared test; h, hours; n, number.

Table 5
Comparison of VAS pain score between two groups.

Different periods	TEAS group (n = 57)	Sham TEAS group (n = 58)	Statistics	P-value
0–2 h after operation	2 (1–3)	3 (2–4)	$z = 1.692$[a]	0.091
2–6 h after operation	2 (1–3)	3 (2–3.25)	$z = 1.900$[a]	0.057
6–24 h after operation	2 (0.5–3)	4 (3–4)	$z = 4.007$[a]	0.001

VAS, visual analog scale; TEAS, transcutaneous electrical acupoint stimulation.
[a] Mann–Whitney *U* test; h, hours; n, number.

and 6–24 h following craniotomy; however, it reduced vomiting incidence at 2–6 h following craniotomy; this finding may be related to the highest incidence of nausea and vomiting observed at 2–6 h following craniotomy. In a previous study, the highest incidence of vomiting was observed at 2–6 h following craniotomy [17], which is similar to that observed in the present study. In addition, in the present study, the vomiting incidence was 29.3 % in the sham TEAS group and 14.0 % in the TEAS group at 2–6 h following craniotomy. However, the vomiting incidence at 6–24 h following craniotomy was still lower in the TEAS group than in the sham TEAS group, although the difference was not statistically significant. These results are consistent with the findings of previous studies [9,18,19]. Moreover, a previous meta-analysis [20] reported that TEAS can be reasonably incorporated into a multimodal management approach to prevent PONV. Our findings were inconsistent with those of Liu et al. who found no difference in the incidence and severity of PONV after using TEAS [12]. We hypothesized that this lack of effect could be due to the different acupoints used in TEAS. They used Hegu (LI4) and Waiguan (TE5), Jinmen (BL63) and Taichong (LR3), Zusanli (ST36) and Qiuxu (GB 40), Fengchi (GB20) and Tianzhu (BL10), and Cuanzhu (BL2) and Yuyao (EX-HN4) on the craniotomy side; however, these acupoints are not usually used to treat nausea and vomiting [12]. Many acupoints, including the P6 site, have been considerably investigated for their postoperative antiemetic effects and have demonstrated efficacy in other surgeries [21].

The present study found that TEAS at P6 did not reduce the degree of nausea in patients after craniotomy, which is consistent with the results of a previous study [22]. However, some previous studies have reported contrasting results, i.e., they have reported that the degree of nausea decreased after TEAS [23,24]. This discrepancy in the results may be attributed to the inclusion of different participants. The abovementioned studies included patients who underwent gynecological surgery, and all of them were female, whereas in the present study, we included participants who underwent craniotomy, with both males and females being included. As we know, sex is an independent risk factor for the incidence and extent of PONV [4]. TEAS at P6 did not reduce the dosage of antiemetic drugs administered to patients following craniotomy; however, we found that the proportion of patients reusing antiemetic drugs was low; in fact, only one patient in our study required metoclopramide administration within 24 h after craniotomy.

Intraoperative and postoperative analgesia drug use can affect the incidence of PONV. In the present study, no significant difference in intraoperative analgesic dosage was observed between the two groups. At 6 h after craniotomy, if the VAS pain score of the patient was >6, 10 mg tramadol was administered as advised by a neurosurgeon. The proportion of patients who received tramadol was compared between the two groups, and no difference was observed between the two groups at 6–24 h following craniotomy.

In the present study, no statistically significant difference in the pain scores was observed between the two groups at 0–2 and 2–6 h following craniotomy. We speculate that this result was related to the plasma half-life of intraoperative analgesics. After 6 h, the analgesic effect of a drug diminishes because of drug metabolism. In contrast, the pain scores were lower in the TEAS group than in the sham TEAS group at 6–24 h after craniotomy. These results indicated that TEAS at P6 reduced the postoperative pain at 6–24 h following craniotomy. This result was consistent with that of Liu et al. [12]. They found that TEAS may reduce pain scores following craniotomy on postoperative day 1; nevertheless, they did not focus on different time points in a 24-h period. In addition, previous studies have demonstrated that TEAS can reduce postoperative pain following gastric and colorectal surgeries [9,25].

The antiemetic effect of acupuncture is the result of increased endorphin and adrenocortical hormone production in the brain, which inhibits the vomiting center through the chemoreceptor-triggering areas [26]. Although research on PONV prevention and treatment based on P6 stimulation has progressed, the specific physiological and biochemical mechanisms that occur following craniotomy need to be further investigated.

Our study has two limitations. First, we did not investigate the mechanism by which TEAS at P6 prevents vomiting following craniotomy. Second, we conducted only one session of stimulation and did not compare multiple stimulations following craniotomy. In the future, we hope to conduct a multicenter study involving more ethnic groups to validate the findings of the present study.

5. Conclusion

TEAS at P6 may reduce vomiting incidence and pain scores following craniotomy; however, the underlying mechanism must be elucidated in the future.

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Data availability

The datasets used and/or analyzed are available from the corresponding author upon reasonable request.

Declaration of competing interest

The authors declare no conflict of interest.

CRediT authorship contribution statement

Liang-dan Tu: Data curation, Conceptualization. **Peng-cheng Li:** Investigation. **Yu Zhao:** Methodology. **Rui-zhi Feng:** Formal analysis, Data curation. **Jian-qin Lv:** Writing – original draft, Project administration, Conceptualization.

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